CSCI250 Computer Organization

Chapter 1 Part 2

September 2, 2023

1 Optimizing Performance
- Optimizing Program Performance
- Latency
- CPU Performance

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Optimizing Program Performance - 1

- Programmers have long been concerned about the performance of their programs
- During the 1960's and 1970's memory capacity was the primary constraint on computer performance.
  - Programmers had to write small fast code to fit into available memory spaces typically measured in Kb (1000s of bytes).
  - Often had to code in assembly language to directly control how much memory was used by the computer program code.

Optimizing Program Performance - 2

- Today personal computers commonly have gigabytes of memory
  - program size is less critical for software performance
  - consider the amount of disk space required to store complete programs for common applications such as Microsoft Office.
  - How much space is used by Microsoft Office 2010 (what I have)
    - 889 MB (Office 365 takes 3 GB).
  - How much space is used by Windows 10 (64 bit)?
    - 20 GB.

Optimizing Program Performance - 3

- Can you think of current computing platforms where program memory size is a concern? How much memory is available?
  - Portable game consoles?
  - Telephones?

Optimizing Program Performance - 4

- Programmers should understand as much as possible about the design and performance characteristics of current computing hardware including the memory, processor, buses, and I/O devices.
- This knowledge will enable the programmer to be better able to design an efficient and optimized piece of software.

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Relative Processing Latencies

- a 1 GHz processor may complete one instruction in 1 billionth of a second (1 ns)
- a fast RAM memory may satisfy a read request in 50 ns
- a fast hard drive may satisfy a read request in 5 ms (5/1000 sec)
- what is the order of magnitude (factor of 10) difference between each of these times?

Implications of Relative Latencies

- Question: What lessons should a programmer learn from these order of magnitude differences between the latencies of processor, RAM, and hard disk in order to make a program run as fast as possible?
  - Minimize reads/writes to disk storage – it's slow
  - Choose sequences of machine instructions that consume the least number of cycles
    - Example if ADD is 2 cycles and MUL is 6 cycles
    - Compute Y = 2 * X as Y = X + X instead
  - Later, we learn how caches can minimize the impact due to these drastically different latencies

Optimizing Program Performance - 5

- Some key questions
  - How much program data can be accessed from the processor's cache memory thus giving the absolute fastest performance?
  - How much slower is the program operation if data must be accessed from main memory?
  - How much slower still would the program operation be if the data had to be accessed from the hard disk?

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Execution Time Equations

$CPUExecutionTime =$

- $CPUClockCyclesForAProgram * ClockCycleTime$

Alternatively

- $\frac{CPUClockCyclesForAProgram}{ClockRate}$

Some Relationships

Clock rate is the reciprocal of clock cycle time

- $cycleTime = \frac{1}{clockRate}$

Execution time is the reciprocal of performance

- $Performance = \frac{1}{executionTime}$

So if performanceA >performanceB ==

- $\frac{1}{executionTimeA} > \frac{1}{executionTimeB}$ ::
- executionTimeB >executionTimeA

Instruction Performance Equations

Clock Cycles per Instruction (CPI)

- CPU clock cycles =
  instructions for a program
  * average clock cycles per instruction

Problem Example

Suppose we have two implementations of the same instruction set architecture. Computer A has a clock cycle time of 250 ps and a CPI of 2.0 for some program, and computer B has a clock cycle time of 500 ps and a CPI of 1.2 for the same program.
Which computer is faster for this program and by how much?

Solution Step-by-Step

We were given the clock cycle time and CPI for two programs. We don't have the number of instructions, but we don't need the exact numbers for this ratio.

- CPU clock $cycles_A = I * 2.0$
- CPU clock $cycles_B = I * 1.2$
- CPU $time_A = I * 2.0 * 250$ ps $= 500 * I$ps
- CPU $time_B = I * 1.2 * 500$ ps $= 600 * I$ps
- So A is faster, but by how much?
- $\frac{CPUPerformance_A}{CPUPerformance_B} = \frac{ExecutionTime_B}{ExecutionTime_A} = \frac{600*Ips}{500*Ips} = 1.2$ times faster

The Classic CPU Performance Equation

- CPU time = instructionCount * CPI * clockCycleTime

OR

- CPU time = $\frac{instructionCount * CPI}{clockRate}$

Use the one you can with the information you are given.

Comparing Code Segments - Example

A compiler designer is trying to decide between two code sequences for a particular computer. They have given us this information:

	CPI for each instruction class		
	A	B	C
CPI	1	2	3

For a particular high-level language statement, the compiler writer is considering two code sequences that require the following instruction counts.

	Instruction counts for each instruction class		
Code sequence	A	B	C
1	2	1	2
2	4	1	1

Comparing Code Segments - Example cont.

- Which code sequence executes the most instructions?
- Which will be faster?
- What is the CPI for each sequence?

Solution

- $\text{CPUclockCycles} = \sum_{i=1}^{n}(CPI_i * C_i)$
- $CPUclockCycles_1 = (1*2)+(2*1)+(3*2) = 2+2+6 = 10$ cycles
- $CPUclockCycles_2 = (1*4)+(2*1)+(3*1) = 4+2+3 = 9$ cycles
- $CPI_1 = \frac{CPUclockCycles_1}{InstructionCount_1} = \frac{10}{5} = 2.0$
- $CPI_2 = \frac{CPUclockCycles_2}{InstructionCount_2} = \frac{9}{6} = 1.5$

Note C stands for the instruction counts for each class

Components of Performance

Components of performance	Units of Measure
CPU execution time for a program	seconds for the program to execute
Instruction Count	Instructions executed for the program
Clock cycles per instruction (CPI)	Average number of clock cycles per instruction
Clock Cycle Time	Seconds per clock cycle

Understanding Program Performance

Hardware or software component	Affects What?	How?
algorithm	instruction count and possibly CPI	Algorithm determines instruction count. Data types affect CPI (floating points are higher)
programming language	instruction count, CPI	Programming language determines instruction count and language features affect CPI
compiler	instruction count, CPI	The compiler determines the translation of language instruction into computer instructions
Instruction Set Architecture	instruction count, clock rate, CPI	Affects the instructions needed for a function, the cost of cycles for each instruction and the clock rate of the processor

No Lab this Lecture

Download the handout of Performance Equations.
Download the worksheet for the first quiz.
Work your way through the problems.
Compare answers with your classmates and when you are sure you have the right answers, take the quiz by typing the answers from the worksheet into the quiz.